

Mathematics Class Notes: Differential Calculus & Mean Value Theorems

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1 Definitions: Continuity & Differentiability

Continuity at a Point

Definition: A function $f(x)$ is said to be continuous at a point $x = a$ if for every small $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x :

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$$

Differentiability at a Point

Definition: A function $f(x)$ is differentiable at a point $x = a$ if the limit of the derivative exists:

$$\lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \text{ exists.}$$

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2 Rolle's Theorem

Statement

If a real-valued function $f(x)$ satisfies:

1. $f(x)$ is continuous on the closed interval $[a, b]$
2. $f(x)$ is differentiable on the open interval (a, b)
3. $f(a) = f(b)$

Then there exists at least one value $c \in (a, b)$ such that:

$$f'(c) = 0$$

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Worked Examples for Rolle's Theorem

Example 1: Verify Rolle's Theorem for $f(x) = x(x+3)e^{-x/2}$ on $[-3, 0]$.

Solution:

$$f(-3) = 0, \quad f(0) = 0 \implies f(-3) = f(0)$$

Differentiating $f(x)$:

$$f'(x) = (2x+3)e^{-x/2} + x(x+3)e^{-x/2} \left(-\frac{1}{2}\right) = \frac{-x^2 + x + 6}{2}e^{-x/2}$$

Set $f'(c) = 0$:

$$-c^2 + c + 6 = 0 \implies c^2 - c - 6 = 0 \implies (c-3)(c+2) = 0$$

$$c = -2 \quad \text{or} \quad c = 3$$

Since $c = -2 \in (-3, 0)$, Rolle's Theorem is verified for $c = -2$.

Example 2: Show that $f(x) = x^5 + 5x + 10 = 0$ has exactly one real root.

Solution:

Assume $f(x) = 0$ has two real roots, say a and b . Then $f(a) = f(b) = 0$. By Rolle's Theorem, there exists $c \in (a, b)$ such that $f'(c) = 0$. Calculated derivative:

$$f'(x) = 5x^4 + 10$$

Set $f'(c) = 5c^4 + 10 = 0 \implies c^4 = -2$, which has no real solutions. This contradiction proves that $f(x) = 0$ cannot have two real roots; hence, it has ****exactly one real root****.

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3 Lagrange's Mean Value Theorem (LMVT)

Statement

If a function $f(x)$ is:

1. Continuous on the closed interval $[a, b]$
2. Differentiable on the open interval (a, b)

Then there exists at least one value $c \in (a, b)$ such that:

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

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Worked Example for LMVT

Example: Find c using LMVT for $f(x) = x(x-1)(x-2)$ on $[0, 1/2]$.

Solution:

$$f(x) = x^3 - 3x^2 + 2x \implies f'(x) = 3x^2 - 6x + 2$$

Evaluate at boundary points:

$$f(0) = 0, \quad f\left(\frac{1}{2}\right) = \frac{1}{2} \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) = \frac{3}{8}$$

Apply LMVT:

$$\frac{f(1/2) - f(0)}{\frac{1}{2} - 0} = \frac{3/8}{1/2} = \frac{3}{4}$$

Set $f'(c) = \frac{3}{4}$:

$$3c^2 - 6c + 2 = \frac{3}{4} \implies 3c^2 - 6c + \frac{5}{4} = 0 \implies 12c^2 - 24c + 5 = 0$$

Solving for c using the quadratic formula:

$$c = \frac{24 \pm \sqrt{576 - 240}}{24} = \frac{6 \pm \sqrt{21}}{6}$$

Choosing the value within $(0, 1/2)$:

$$c = \frac{6 - \sqrt{21}}{6} \approx 0.236 \in \left(0, \frac{1}{2}\right)$$

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4 Cauchy's Mean Value Theorem (CMVT)

Statement

If two functions $f(x)$ and $g(x)$ are:

1. Continuous on $[a, b]$
2. Differentiable on (a, b)
3. $g'(x) \neq 0$ for all $x \in (a, b)$

Then there exists at least one value $c \in (a, b)$ such that:

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

—

Worked Examples for CMVT

Example 1: Verify CMVT for $f(x) = \ln x$ and $g(x) = \frac{1}{x}$ on $[1, e]$.

Solution:

$$f'(x) = \frac{1}{x}, \quad g'(x) = -\frac{1}{x^2} \implies \frac{f'(c)}{g'(c)} = \frac{1/c}{-1/c^2} = -c$$

Evaluate boundary values:

$$\frac{f(e) - f(1)}{g(e) - g(1)} = \frac{\ln e - \ln 1}{\frac{1}{e} - 1} = \frac{1 - 0}{\frac{1-e}{e}} = \frac{e}{1-e}$$

Set $-c = \frac{e}{1-e}$:

$$c = \frac{e}{e-1} \in (1, e)$$

Example 2: Show that if $f(x) = e^x$ and $g(x) = e^{-x}$, then c is the arithmetic mean of a and b .

Solution:

$$f'(x) = e^x, \quad g'(x) = -e^{-x} \implies \frac{f'(c)}{g'(c)} = \frac{e^c}{-e^{-c}} = -e^{2c}$$

Apply CMVT:

$$\frac{e^b - e^a}{e^{-b} - e^{-a}} = \frac{e^b - e^a}{\frac{e^a - e^b}{e^a e^b}} = -e^{a+b}$$

Equating both sides:

$$-e^{2c} = -e^{a+b} \implies 2c = a + b \implies c = \frac{a+b}{2}$$

Example 3: Show $c = \sqrt{ab}$ (geometric mean) using CMVT for $f(x) = \frac{1}{\sqrt{x}}$ and $g(x) = \frac{1}{x^2}$.

Solution:

$$f'(x) = -\frac{1}{2}x^{-3/2}, \quad g'(x) = -\frac{2}{x^3}$$
$$\frac{f'(c)}{g'(c)} = \frac{-\frac{1}{2}c^{-3/2}}{-2c^{-3}} = \frac{1}{4}c^{3/2}$$

Applying boundary evaluations yields:

$$c = \sqrt{ab}$$

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5 Implications and Applications of Mean Value Theorems

Key Monotonicity Properties

1. If $f'(x) > 0$ for all x , then $f(x)$ is ****strictly increasing****.
2. If $f'(x) < 0$ for all x , then $f(x)$ is ****strictly decreasing****.
3. If $f'(x) = 0$ for all x , then $f(x)$ is ****constant****.

Inequality Example 1: Show $1 + x < e^x < 1 + xe^x$ for $x > 0$.

Proof:

Let $f(x) = e^x - 1 - x$.

$f'(x) = e^x - 1 > 0$ for all $x > 0$.

Since $f(0) = 0$, $f(x) > f(0) \implies e^x - 1 - x > 0 \implies e^x > 1 + x$.

Next, let $g(x) = 1 + xe^x - e^x$.

$g'(x) = e^x + xe^x - e^x = xe^x > 0$ for $x > 0$.

Since $g(0) = 0$, $g(x) > g(0) \implies 1 + xe^x > e^x$.

Combining both inequalities yields:

$$1 + x < e^x < 1 + xe^x \quad \forall x > 0$$

Inequality Example 2: Show $\frac{x}{1+x} < \ln(1+x) < x$ for $x > 0$.

Proof:

Let $f(x) = \ln(1+x) - \frac{x}{1+x}$.

$$f'(x) = \frac{1}{1+x} - \frac{(1+x) - x}{(1+x)^2} = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} > 0 \quad \forall x > 0$$

Since $f(0) = 0$, $f(x) > 0 \implies \ln(1+x) > \frac{x}{1+x}$.

Next, let $g(x) = x - \ln(1+x)$.

$$g'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} > 0 \quad \forall x > 0$$

Since $g(0) = 0$, $g(x) > 0 \implies x > \ln(1+x)$.

Combining both:

$$\frac{x}{1+x} < \ln(1+x) < x \quad \forall x > 0$$

6 Cauchy's Mean Value Theorem (CMVT) & Applications

Cauchy's MVT Applications

Example 1: Show that there exists a number $c \in (a, b)$ such that:

$$2c[f(a) - f(b)] = f'(c)[a^2 - b^2]$$

Proof:

Applying Cauchy's Mean Value Theorem to $f(x)$ and $g(x) = x^2$ on $[a, b]$:

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Since $g'(x) = 2x$, we have $g'(c) = 2c$ and $g(b) - g(a) = b^2 - a^2$:

$$\frac{f(b) - f(a)}{b^2 - a^2} = \frac{f'(c)}{2c} \implies 2c[f(b) - f(a)] = f'(c)[b^2 - a^2]$$

Rearranging terms yields:

$$2c[f(a) - f(b)] = f'(c)[a^2 - b^2]$$

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Example 2: Show that for $0 < a < b < \frac{\pi}{2}$:

$$0 < \cos a - \cos b < b - a$$

Proof:

Let $f(x) = \cos x$ on $[a, b] \subset [0, \frac{\pi}{2}]$.

By LMVT, there exists $c \in (a, b)$ such that:

$$\frac{f(b) - f(a)}{b - a} = f'(c) \implies \cos b - \cos a = -(b - a) \sin c$$

$$\implies \cos a - \cos b = (b - a) \sin c$$

Since $0 < a < c < b < \frac{\pi}{2}$, we have $0 < \sin c < 1$. Thus:

$$0 < \cos a - \cos b < b - a$$

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Example 3: Let f be continuous on $[a, b]$ with $a > 0$, and differentiable on (a, b) . Prove that there exists $c \in (a, b)$ such that:

$$\frac{bf(a) - af(b)}{b - a} = f(c) - cf'(c)$$

Proof:

Apply Cauchy's Mean Value Theorem to $g(x) = \frac{f(x)}{x}$ and $h(x) = \frac{1}{x}$:

$$\frac{g(b) - g(a)}{h(b) - h(a)} = \frac{g'(c)}{h'(c)}$$

Computing derivatives:

$$g'(x) = \frac{xf'(x) - f(x)}{x^2}, \quad h'(x) = -\frac{1}{x^2}$$

Substituting into CMVT:

$$\frac{\frac{f(b)}{b} - \frac{f(a)}{a}}{\frac{1}{b} - \frac{1}{a}} = \frac{\frac{cf'(c) - f(c)}{c^2}}{-\frac{1}{c^2}} = f(c) - cf'(c)$$

Simplifying the left-hand side:

$$\frac{\frac{af(b) - bf(a)}{ab}}{\frac{a-b}{ab}} = \frac{af(b) - bf(a)}{a - b} = \frac{bf(a) - af(b)}{b - a}$$

Therefore:

$$\frac{bf(a) - af(b)}{b - a} = f(c) - cf'(c)$$

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7 Arc Length Differential Formulas

1. Cartesian Form

For a curve given by $y = f(x)$:

$$\left(\frac{ds}{dx}\right)^2 = 1 + \left(\frac{dy}{dx}\right)^2 \implies \frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

2. Parametric Form

For a curve given parametrically by $x = x(t), y = y(t)$:

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

3. Polar Form

For a curve in polar coordinates $r = f(\theta)$:

$$\frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$$

Arc Length Examples

Example 1 (Parametric): Find the perimeter of the circle $x = \cos t, y = \sin t$ for $0 \leq t \leq 2\pi$.

$$\begin{aligned}\frac{dx}{dt} &= -\sin t, & \frac{dy}{dt} &= \cos t \\ \frac{ds}{dt} &= \sqrt{(-\sin t)^2 + (\cos t)^2} = 1 \\ s &= \int_0^{2\pi} 1 \, dt = 2\pi\end{aligned}$$

Example 2 (Cartesian): Find $\frac{ds}{dx}$ for the catenary $y = c \cosh(x/c)$.

$$\begin{aligned}\frac{dy}{dx} &= \sinh(x/c) \\ \frac{ds}{dx} &= \sqrt{1 + \sinh^2(x/c)} = \sqrt{\cosh^2(x/c)} = \cosh(x/c)\end{aligned}$$

Example 3 (Polar): Find $\frac{ds}{d\theta}$ for the cardioid $r = a(1 + \cos \theta)$.

$$\frac{dr}{d\theta} = -a \sin \theta$$

$$\begin{aligned} \frac{ds}{d\theta} &= \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = \sqrt{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta} \\ &= \sqrt{a^2(1 + 2\cos \theta + \cos^2 \theta + \sin^2 \theta)} = \sqrt{2a^2(1 + \cos \theta)} \\ &= \sqrt{2a^2 \cdot 2\cos^2(\theta/2)} = 2a \cos(\theta/2) \end{aligned}$$

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8 Curvature and Radius of Curvature (ρ)

Definitions

- **Curvature (κ):** The rate of change of direction angle ψ with respect to arc length s :

$$\kappa = \frac{d\psi}{ds}$$

- **Radius of Curvature (ρ):** The reciprocal of curvature:

$$\rho = \frac{1}{\kappa} = \frac{ds}{d\psi}$$

Theorem: Prove that the curvature of a circle is constant.

For a circle of radius R , arc length $\Delta s = R\Delta\psi \implies \frac{\Delta\psi}{\Delta s} = \frac{1}{R}$. Taking the limit as $\Delta s \rightarrow 0$:

$$\kappa = \frac{d\psi}{ds} = \frac{1}{R} \quad (\text{Constant})$$

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Radius of Curvature Formulas

1. Cartesian Coordinates

Given $y = f(x)$, let $y_1 = \frac{dy}{dx}$ and $y_2 = \frac{d^2y}{dx^2}$:

$$\tan \psi = y_1 \implies \sec^2 \psi \frac{d\psi}{ds} = y_2 \frac{dx}{ds}$$

$$\rho = \frac{ds}{d\psi} = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

2. Parametric Coordinates

For $x = x(t)$, $y = y(t)$, letting $x' = \frac{dx}{dt}$, $y' = \frac{dy}{dt}$, $x'' = \frac{d^2x}{dt^2}$, $y'' = \frac{d^2y}{dt^2}$:

$$\rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''}$$

3. Polar Coordinates

For $r = f(\theta)$, letting $r_1 = \frac{dr}{d\theta}$ and $r_2 = \frac{d^2r}{d\theta^2}$:

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$$

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Worked Examples for Radius of Curvature

Example 1 (Cartesian): Find the radius of curvature ρ for the parabola $y^2 = 8x$ at any point.

Solution:

$$2yy_1 = 8 \implies y_1 = \frac{4}{y} = \frac{4}{2\sqrt{2}x^{1/2}} = \frac{\sqrt{2}}{x^{1/2}}$$
$$y_2 = \frac{d}{dx} \left(\sqrt{2}x^{-1/2} \right) = -\frac{1}{\sqrt{2}}x^{-3/2}$$

Substitute into Cartesian formula:

$$\rho = \frac{\left(1 + \frac{2}{x}\right)^{3/2}}{-\frac{1}{\sqrt{2}}x^{-3/2}} = -\sqrt{2}(x+2)^{3/2}$$

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Example 2 (Parametric/Polar): Find the radius of curvature at any point on the cycloid:

$$x = a(\theta + \sin \theta), \quad y = a(1 - \cos \theta)$$

Solution:

$$x' = a(1 + \cos \theta), \quad y' = a \sin \theta$$
$$x'' = -a \sin \theta, \quad y'' = a \cos \theta$$

Calculate $x'^2 + y'^2$:

$$x'^2 + y'^2 = a^2(1 + 2\cos \theta + \cos^2 \theta + \sin^2 \theta) = 2a^2(1 + \cos \theta)$$

Calculate denominator $x'y'' - y'x''$:

$$a(1 + \cos \theta)(a \cos \theta) - (a \sin \theta)(-a \sin \theta) = a^2(\cos \theta + \cos^2 \theta + \sin^2 \theta) = a^2(1 + \cos \theta)$$

Substitute into parametric formula:

$$\rho = \frac{[2a^2(1 + \cos \theta)]^{3/2}}{a^2(1 + \cos \theta)} = 2\sqrt{2}a\sqrt{1 + \cos \theta}$$

Using $\sqrt{1 + \cos \theta} = \sqrt{2} \cos(\theta/2)$:

$$\rho = 2\sqrt{2}a \cdot \sqrt{2} \cos \left(\frac{\theta}{2} \right) = 4a \cos \left(\frac{\theta}{2} \right)$$

9 Newton's Method for Radius of Curvature at the Origin

Formulae

1. If x -axis is the tangent to the curve at the origin:

$$\rho = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \left(\frac{x^2}{2y} \right)$$

2. If y -axis is the tangent to the curve at the origin:

$$\rho = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \left(\frac{y^2}{2x} \right)$$

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Examples

Example 1: Find the radius of curvature at the origin for the curve:

$$2x^4 + 4x^3y + xy^2 + 6y^3 - 3x^2 + 2xy + y^2 - 4x = 0$$

Solution:

At $x = 0, y = 0$, the equation is satisfied $\implies$ the curve passes through the origin.

To find the tangent at the origin, equate the lowest degree terms to zero:

$$-4x = 0 \implies x = 0 \quad (\text{y-axis is the tangent at the origin})$$

Using Newton's formula for y -axis tangent:

$$\rho = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \left(\frac{y^2}{2x} \right)$$

Divide the original equation by $2x$:

$$x^3 + 2x^2y + \frac{y^2}{2} + \frac{6y^3}{2x} - \frac{3x}{2} + y + \frac{y^2}{2x} - 2 = 0$$

Re-writing $\frac{6y^3}{2x}$ as $6y \cdot \left(\frac{y^2}{2x}\right)$:

$$x^3 + 2x^2y + \frac{y^2}{2} + 6y \left(\frac{y^2}{2x}\right) - \frac{3x}{2} + y + \left(\frac{y^2}{2x}\right) - 2 = 0$$

Taking the limit as $x \rightarrow 0, y \rightarrow 0$ and substituting $\lim \frac{y^2}{2x} = \rho$:

$$0 + 0 + 0 + 6(0)\rho - 0 + 0 + \rho - 2 = 0$$

$$\implies \rho - 2 = 0 \implies \rho = 2$$

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Example 2: Find the radius of curvature at the origin for the curve:

$$x^3 - 2x^2y + 3xy^2 - 4y^3 + 5x^2 - 6xy + 7y^2 - 8y = 0$$

Solution:

Equating the lowest degree terms to zero:

$$-8y = 0 \implies y = 0 \quad (\text{x-axis is the tangent at the origin})$$

Divide the equation by $2y$:

$$\frac{x^3}{2y} - x^2 + \frac{3}{2}xy - 2y^2 + \frac{5x^2}{2y} - 3x + \frac{7}{2}y - 4 = 0$$

Taking the limit as $x \rightarrow 0, y \rightarrow 0$ and noting $\lim \frac{x^2}{2y} = \rho$:

$$0 \cdot \rho - 0 + 0 - 0 + 5\rho - 0 + 0 - 4 = 0$$

$$\implies 5\rho - 4 = 0 \implies \rho = \frac{4}{5}$$

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Example 3: Find the radius of curvature at the origin for:

$$x^3y - xy^3 + 2x^2y + xy - y^2 + 2x = 0$$

Solution:

Equating lowest degree terms:

$$2x = 0 \implies x = 0 \quad (\text{y-axis is the tangent at the origin})$$

Dividing equation by $2x$:

$$x^2 \left(\frac{y}{2x} \right) - xy^2 \left(\frac{y}{2x} \right) + 2x^2 \left(\frac{y}{2x} \right) + x \left(\frac{y}{2x} \right) - y \left(\frac{y}{2x} \right) + 1 = 0$$

Taking limits $x \rightarrow 0, y \rightarrow 0$ with $\lim \frac{y^2}{2x} = \rho$:

$$0 - 0 + 0 + 0 - \rho + 1 = 0 \implies -\rho + 1 = 0 \implies \rho = 1$$

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10 Asymptotes of Algebraic Curves

Horizontal and Vertical Asymptotes

- **Vertical Asymptote:** Occurs where $f(x) \rightarrow \pm\infty$ as $x \rightarrow a$.
- **Horizontal Asymptote:** $y = b$ where $\lim_{x \rightarrow \pm\infty} f(x) = b$.

Example: Find asymptotes of $f(x) = \frac{x^2-x-2}{x-3}$.

- Vertical asymptote: Set denominator to zero $\implies x - 3 = 0 \implies \mathbf{x = 3}$.
- Oblique asymptote: Divide numerator by denominator:

$$f(x) = x + 2 + \frac{4}{x-3} \implies \mathbf{y = x + 2}$$

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Oblique Asymptotes ($y = mx + c$)

For an n -th degree curve:

$$\phi_n(m) = 0 \implies \text{Solve for } m$$

Then c is determined by:

1. If $\phi'_n(m) \neq 0$:

$$c = -\frac{\phi_{n-1}(m)}{\phi'_n(m)}$$

2. If $\phi'_n(m) = 0$ and $\phi_{n-1}(m) = 0$:

$$\frac{c^2}{2!}\phi''_n(m) + c\phi'_{n-1}(m) + \phi_{n-2}(m) = 0$$

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Worked Examples for Asymptotes

Example 1: Find the asymptotes of $xy^2 - x^2y - 3x^2 - 2xy + y^2 + x - 2y + 1 = 0$.

Solution:

Degree $n = 3$.

Highest degree terms ($n = 3$): $\phi_3(m) = m^2 - m = m(m-1) \implies m = 0, m = 1$. Degree $n = 2$ terms: $\phi_2(m) = -3 - 2m + m^2$.

- For $m = 0$:

$$\begin{aligned}\phi'_3(m) &= 2m - 1 \implies \phi'_3(0) = -1 \\ \phi_2(0) &= -3 \implies c = -\frac{-3}{-1} = -3 \implies \mathbf{y = -3}\end{aligned}$$

- For $m = 1$:

$$\begin{aligned}\phi'_3(1) &= 2(1) - 1 = 1 \\ \phi_2(1) &= -3 - 2 + 1 = -4 \implies c = -\frac{-4}{1} = 4 \implies \mathbf{y = x + 4}\end{aligned}$$

—

Example 2: Find asymptotes of $x^3 - x^2y - xy^2 + y^3 + 2x^2 - 4y^2 + 2xy + x + y + 1 = 0$.

Solution:

$$\phi_3(m) = 1 - m - m^2 + m^3 = (1 - m)(1 - m^2) = (1 - m)^2(1 + m) = 0 \implies m = 1, 1, -1.$$

- For non-repeated root $m = -1$:

$$\phi'_3(m) = -1 - 2m + 3m^2 \implies \phi'_3(-1) = -1 + 2 + 3 = 4$$

$$\phi_2(m) = 2 - 4m^2 + 2m \implies \phi_2(-1) = 2 - 4 - 2 = -4$$

$$c = -\frac{\phi_2(-1)}{\phi'_3(-1)} = -\frac{-4}{4} = 1 \implies \mathbf{y} = -\mathbf{x} + 1$$

- For repeated roots $m = 1, 1$:

Since $\phi'_3(1) = 0$, use quadratic in c :

$$\frac{c^2}{2}\phi''_3(m) + c\phi'_2(m) + \phi_1(m) = 0$$

Evaluating derivatives at $m = 1$:

$$\phi''_3(1) = -2 + 6(1) = 4, \quad \phi'_2(1) = -8(1) + 2 = -6, \quad \phi_1(1) = 1 + 1 = 2$$

$$\frac{c^2}{2}(4) + c(-6) + 2 = 0 \implies 2c^2 - 6c + 2 = 0 \implies c^2 - 3c + 1 = 0$$

$$c = \frac{3 \pm \sqrt{9 - 4}}{2} = \frac{3 \pm \sqrt{5}}{2}$$

Asymptotes: $\mathbf{y} = \mathbf{x} + \frac{3 \pm \sqrt{5}}{2}$.

11 Intersection of Curve and Asymptotes

Theorems

1. An n -th degree curve has n asymptotes.
2. Each asymptote cuts the curve at most in $(n - 2)$ points.
3. All n asymptotes together cut the curve in at most $n(n - 2)$ points.
4. If the equation of an n -th degree curve can be written as:

$$F_n + F_{n-2} = 0$$

where $F_n = 0$ represents the product of n asymptotes, then $F_{n-2} = 0$ is the equation of the curve passing through the $n(n - 2)$ points of intersection.

Examples

Example 1: Show that the asymptotes of the curve:

$$x^3 - xy^2 + 2xy + 2x - y = 0$$

cut the curve again at 3 points lying on a straight line.

Solution:

Here $n = 3$. Maximum intersection points $= 3(3 - 2) = 3$.

Factorizing the highest degree terms gives asymptotes:

$$(x - y - 1)(x + y + 1) = 0 \quad \text{and} \quad x = 0$$

The curve equation rearranges to:

$$x(x - y - 1)(x + y + 1) + (3x - y) = 0 \implies F_3 + F_1 = 0$$

The 3 intersection points lie on the line $F_1 = 0$, which is $3x - y = 0$.

Example 2: Show that the asymptotes of $x^3 - x^2y - xy^2 + y^3 + 2x^2 - 4y^2 + 2xy + x + y + 1 = 0$ cut the curve in 3 points lying on a circle.

Solution:

The product of asymptotes is given by $F_3 = 0$.

Rewriting the curve as $F_3 + F_1 = 0$:

$$xy(x - y)(x + y) + \lambda(x^2 + y^2 - a^2) = 0$$

Since the remaining term F_1 or F_2 represents a circle equation $x^2 + y^2 = a^2$, all points of intersection lie on the circle centered at the origin with radius a .